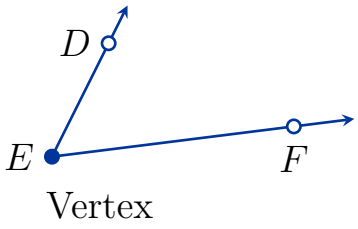
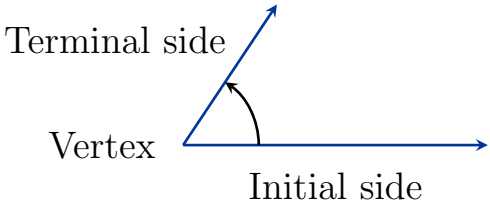
5.1: Angles

Definition.
A **ray** consists of one point on a line, and all points extending in one direction from that point. The following is ray $\overrightarrow{EF}$:

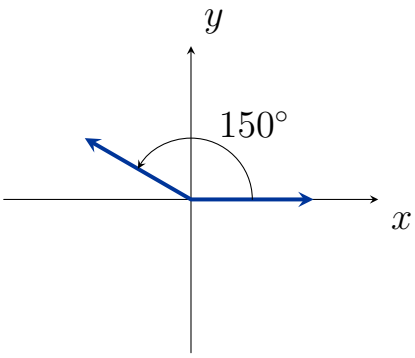


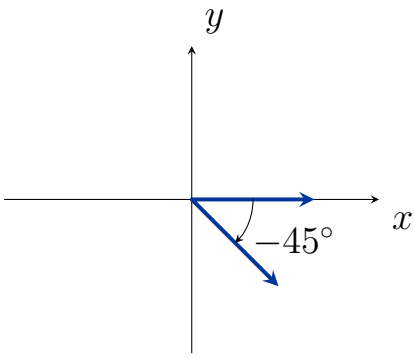
An **angle** is the union of two rays having a common endpoint. The endpoint is called the **vertex** of the angle. Angles can be named using a point on each ray and the vertex. The following is $\angle DEF$:





When we create an angle, we fix one ray, called the **initial side**, and rotate the **terminal side** about the vertex.
An angle drawn on the xy-plane is said to be in **standard position** if its vertex is located at the origin, and the initial side extends along the x-axis:

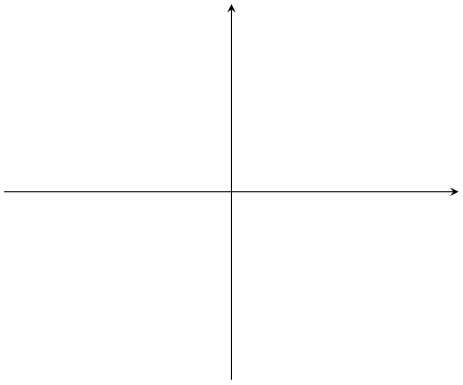




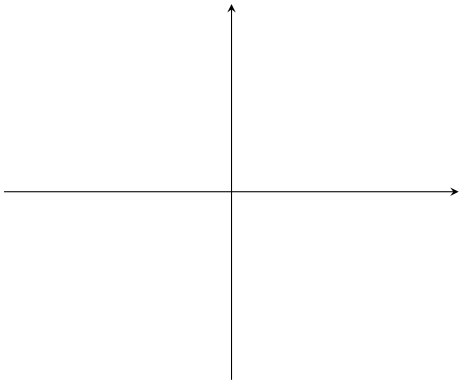
A **positive angle** rotates in the counterclockwise direction. A **negative angle** rotates in the clockwise direction. A full circle rotation is defined to be 360°.

Example. Sketch the following angles in standard position:

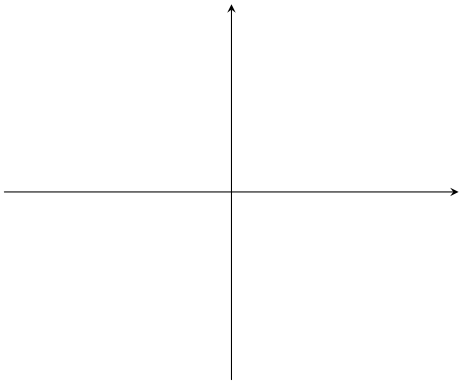
$0°$



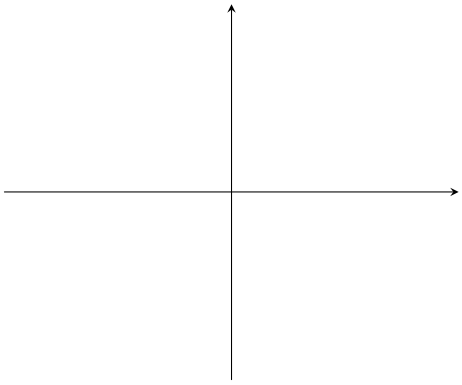
$30°$



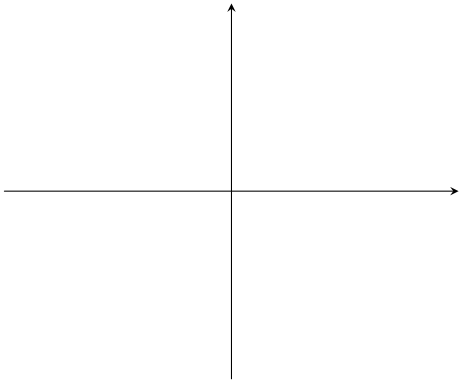
$90°$



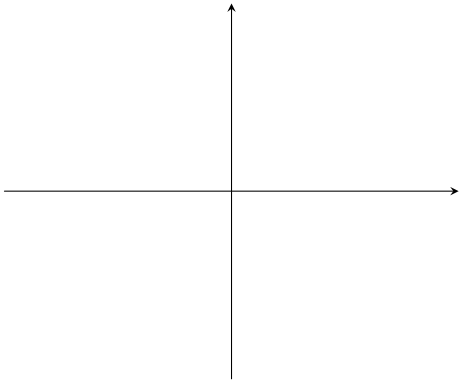
$45°$



$135°$



$-60°$



Definition.
A **radian** is the measure of a central angle corresponding to an arc equal in length to the radius of that circle. Since the circumference of a circle is $C = 2\pi r$, there are 2π radians in a full circle. This gives us

$$2\pi = 360^\circ$$

or, equivalently,

$$\pi = 180^\circ$$

Example. Convert the following angles to radians:

225°

270°

-135°

-45°

Example. Convert the following angles to degrees:

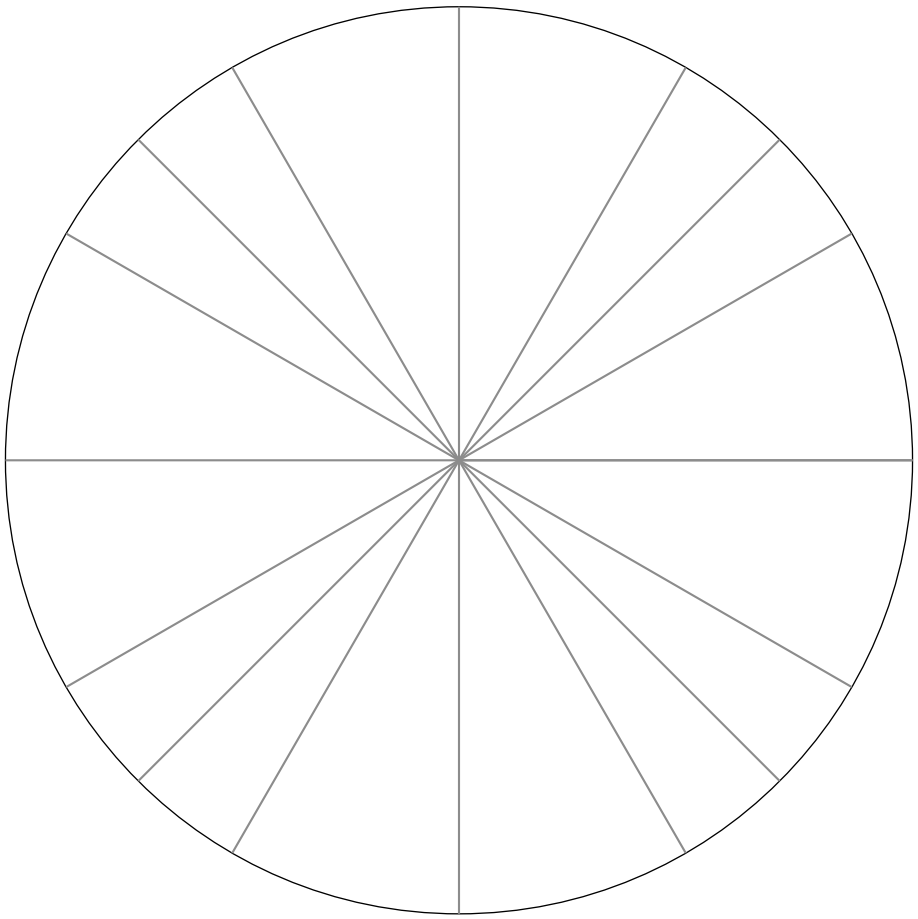
4π

$-\frac{\pi}{2}$

$\frac{\pi}{12}$

$-\frac{5\pi}{4}$

Example. The following circle contains certain angles of interest. Find the angles of each in both degrees and radians:



Definition.
Coterminal angles are two angles in standard position that have the same terminal side.

Example. Find a coterminal angle for each of the following:

400°

-30°

1655°

-500°

7π

$\dfrac{-5\pi}{6}$

$\dfrac{21\pi}{4}$

$-\dfrac{17\pi}{3}$